

Iker Garcia Morales

| +1-479-599-8384 | garciamorai@jbu.edu | [linkedin.com/in/ikermora](https://www.linkedin.com/in/ikermora) | github.com/ikeermora |

Electrical Engineering & Robotics student focused on VLSI, digital design, RTL, hardware verification, and CPU architecture.
Founder and lead developer of Simple Silicon, a desktop EDA environment for SystemVerilog design, simulation, waveform inspection, and RTL visualization.

SKILLS:

HDLs: SystemVerilog, Verilog, VHDL

Digital Design: RTL Design, FSMs, ISA Design, Multicycle Datapaths, Digital Logic

VLSI / ASIC: CMOS Schematic & Layout, DRC, ERC, NCC, ASIC Design Flow

Tools: Quartus Prime, ModelSim, Yosys, Electric VLSI, LTspice, Git, Linux

Programming: Python, C/C++, MATLAB

Embedded: STM32, ESP32, Arduino

EDUCATION:

John Brown University, Siloam Springs, AR

B.S. in Electrical Engineering; Double Major in Robotics and Mechatronics Engineering

Expected May 2027

Relevant Coursework

Digital IC Design, Digital Electronics, Embedded Systems, Electronics, Electrical Circuits, Control Systems.

PROJECTS:

Simple Silicon — Founder & Lead Developer (TypeScript, React, Rust/Tauri, Yosys)

Developing a desktop EDA environment for SystemVerilog design, integrating RTL editing, simulation, waveform inspection, and interactive hardware visualization. Built an RTL Visualizer that uses Yosys to elaborate designs and display module hierarchy, internal logic, gates, multiplexers, signals, and connections. Designed the product architecture, interface, and verification workflows with a focus on making digital hardware development more accessible and intuitive.

Silicio-16: Custom 16-bit Processor (SystemVerilog)

Designed and verified a custom 16-bit multicycle processor in SystemVerilog featuring eight general-purpose registers, a 16-instruction ISA, and support for arithmetic, logic, memory, comparison, branch, and jump operations. Implemented the complete datapath and FSM-based control unit and validated full-program execution using a self-checking regression suite.

ASIC Design: 8-bit MIPS Processor (Electric VLSI)

Designed an 8-bit MIPS-style processor using a bit-slice architecture, including the datapath, ALU, register structures, and FSM-based control unit. Completed full-custom CMOS schematic and layout, passed DRC, ERC, and NCC verification, simulated the processor using IRSIM, and integrated the core into a MOSIS TinyChip padframe.

Content-Addressable Memory (SystemVerilog)

Implemented a synthesizable 4-entry, 32-bit CAM with synchronous writes, valid-bit tracking, parallel combinational lookup, and match/miss outputs. Developed a self-checking testbench that passed all 10 functional tests.

CanSat World Cup 2026 – UNAM/PEU (PCB Design)

Designed the flight computer PCB integrating sensors, power distribution, radio telemetry, and debugging access. The board transmitted data throughout the full flight with minimal packet loss, helping the team place 14th out of 150 teams.

EXPERIENCE:

John Brown University — Siloam Springs, AR

Teaching Assistant – Digital Electronics

Jan 2025 – Present

Facilitate digital circuit design and troubleshooting labs, support students working with logic gates, flip-flops, multiplexers, and counters, and evaluate assignments for circuit correctness and technical precision.

Learning Coach and Engineering Tutor – Student Support Services

Sep 2024 – Present

Provide one-on-one and small-group support in calculus, electrical circuits, technical problem-solving, and study strategies for engineering students.

LEADERSHIP & AWARDS

IEEE Student Branch – Junior Senator, Communications Officer

Student Government Association – Freshman Senator, Junior Senator.

Walton International Scholarship Scholar.